

SIT720 - Machine Learning

11.1 High Distinction Task - Machine Learning Research

Reproduction and Extension of DASMCC for Multi-Class Cardiovascular Disease Prediction from PTB-XL ECG

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Selected paper: N. Sinha et al., "DASMCC: Data Augmented SMOTE Multi-Class Classifier for Prediction of Cardiovascular Diseases Using Time Series Features," IEEE Access, 2023.

Video presentation link: [INSERT ACCESSIBLE LINK BEFORE SUBMISSION]

Code / ZIP / GitHub link: [INSERT ACCESSIBLE LINK BEFORE SUBMISSION]

Abstract

This report reproduces and critically evaluates the DASMCC cardiovascular-disease classification study of Sinha et al. [1] using PTB-XL ECG data. The original work extracts sixteen time-domain ECG feature types, applies SMOTE to address class imbalance, and compares KNN, Random Forest, Gradient Boosting, CatBoost and XGBoost. In this reproduction, PTB-XL version 1.0.3 was processed with WFDB and NeuroKit2. After restricting the data to a single diagnostic superclass, 16,244 records remained; valid ECG features were extracted from 16,167 records (99.53%). A reproducible 11-beat representation produced 176 predictors per ECG. XGBoost achieved the strongest reproduced result with 69.64% accuracy and 0.8455 weighted multiclass AUC, preserving the original paper's model ranking trend but not its reported 93% accuracy. The discrepancy is analysed in relation to dataset version differences, unspecified lead aggregation, ambiguous sequence equalisation, and missing classifier hyperparameters. Part 2 proposes a robust statistical ECG representation that summarises all valid beats using distributional descriptors. This reduces the feature space from 176 to 98 predictors, cuts training time by 51.94% and prediction time by 24.43%, and retains almost identical AUC (0.8443 versus 0.8455), while slightly improving macro-F1. The proposed method therefore improves compactness and computational efficiency rather than raw accuracy.

1. Introduction

Cardiovascular disease (CVD) remains a major global health problem, and electrocardiography (ECG) provides a non-invasive way to observe cardiac electrical activity. Sinha et al. [1] proposed DASMCC, a multi-class machine-learning framework that uses 12-lead ECG recordings to distinguish Normal ECGs from four diagnostic superclasses: myocardial infarction (MI), hypertrophy (HYP), conduction disturbance (CD), and ST-T change (STTC). The study was designed around a lightweight set of time-domain ECG features so that the resulting classifier could support smart-healthcare and edge/cloud deployment scenarios.

The selected paper positions its contribution within a broad body of CVD-prediction research that includes conventional supervised learning, ensemble learning, ECG feature engineering, and deep-learning approaches [1]. Its research gap is practical as well as methodological: rather than relying on raw high-dimensional signals, it reduces each ECG to a small group of clinically interpretable time-domain features, addresses class imbalance with SMOTE, and compares multiple classifiers under a common multi-class setting. The original study reports XGBoost as the strongest overall classifier, with 93% accuracy, 92% precision, 90% recall and 93% F1 score [1].

The purpose of this report is twofold. Part 1 reproduces the published workflow as closely as possible, documents the assumptions required where the paper is underspecified, and analyses the differences between published and reproduced results. Part 2 then addresses a limitation of the fixed-length feature representation by designing a compact statistical feature representation that uses all valid beats in each ECG and evaluates the resulting efficiency-performance trade-off.

2. Part 1 - Reproduction of Published Research

2.1 Research problem, motivation, SOTA and research gap

The DASMCC problem is a five-class ECG classification task: Normal, MI, HYP, CD and STTC. The motivation is early and automated CVD screening from ECG, while keeping the feature space sufficiently lightweight for practical smart-healthcare deployment [1]. The paper reviews conventional classifiers such as decision trees, Random Forest, KNN and SVM, as well as ensemble and deep-learning approaches. It also notes that CVD prediction has been approached using ECG, demographic variables, electronic health records and other biosignals [1].

The main gap addressed by DASMCC is the need for a multi-class framework that combines interpretable ECG morphology and interval features with imbalance handling and a comparison of several ensemble classifiers. Instead of classifying directly from all 12,000 samples in a 10-second, 12-lead 100 Hz ECG, the paper extracts sixteen time-domain feature types using NeuroKit2 [1], [3]. This is intended to reduce computational burden while retaining physiologically meaningful information.

2.2 Dataset and diagnostic classes

The selected paper uses PTB-XL, a publicly available 12-lead ECG dataset [1], [2]. The paper reports 21,837 ECG records from 18,885 subjects and uses the diagnostic superclasses NORM, MI, STTC, CD and HYP. The reproduction used PTB-XL version 1.0.3, for which the downloaded metadata contained 21,799 records. This difference was retained and reported rather than manually altering the dataset to match the paper.

Diagnostic labels were parsed from the SCP statements and mapped to their diagnostic superclass. Records associated with more than one diagnostic superclass, or with no diagnostic superclass, were excluded to match the paper's single-label design. This produced 16,244 candidate ECGs. Feature extraction succeeded for 16,167 ECGs and failed for 77 records, giving a 99.53% extraction success rate.

Class	Final ECG records	Share of extracted set
NORM	9,067	56.08%
MI	2,516	15.56%
STTC	2,385	14.75%
CD	1,664	10.29%
HYP	535	3.31%

2.3 ECG preprocessing and feature extraction

PTB-XL 100 Hz waveforms were loaded with WFDB [2], [9]. NeuroKit2 [3] was used to clean the signal, detect R peaks and delineate P, Q, S and T peaks. The paper specifies sixteen feature types grouped into peak locations, peak amplitudes and intervals [1].

Feature group	Features
Peak locations	P, Q, R, S and T locations
Peak amplitudes	P, Q, R, S and T amplitudes
Time intervals	PQ, ST, QT, PR, RR and QRS intervals

An initial direct alignment of the delineated peak arrays occasionally produced negative interval values. To prevent invalid beat correspondence, the reproduction matched P and Q peaks occurring before each R peak, S and T peaks occurring after it, and retained only beats satisfying $P < Q < R < S < T$. This removed the negative interval artefacts observed in the preliminary implementation.

The paper states that the sixteen extracted features are stored as variable-length lists, later equalised and flattened, but it does not specify the exact target sequence length or padding strategy [1]. In this reproduction, the number of valid beats per ECG had a mean of 10.91 and a median of 11, so each ECG was represented using 11 beats. Longer sequences were truncated to the first 11 valid beats, while shorter sequences were padded by repeating the final valid beat. The resulting model input therefore contained $11 \times 16 = 176$ predictors per ECG. Final-beat RR values were forward/back-filled within the ECG, consistent with the paper's use of forward filling for missing extracted features [1].

2.4 Reproduction assumptions and reproducibility constraints

Several implementation decisions were necessary because the publication does not provide enough detail to reconstruct every step exactly. First, although the paper describes 12-lead ECG input, it does not clearly explain how per-lead extracted features are consolidated into the final flattened representation. Lead II was therefore used as the primary morphology lead for the main reproduction. Second, the paper does not specify the target sequence length used when equalising variable-length feature lists, so the observed median of 11 valid beats was selected. Third, classifier hyperparameters are not fully reported, so explicit reproducible settings were used and documented in code.

The source paper also contains internal ambiguities that affect exact reproducibility. It reports a waveform array of 1000×12 samples but elsewhere describes the raw ECG as having 21,000 dimensions; mathematically, the stated

array contains 12,000 signal values. The abstract also refers to "Adda Boost" while the main modelling and results sections use CatBoost [1]. These issues were treated as reproducibility limitations rather than silently corrected.

2.5 Experimental protocol

The final dataset was split using an 80/20 stratified train-test split with `random_state = 42`. `StandardScaler` and `SMOTE` were placed inside an imbalanced-learn pipeline so that scaling and synthetic oversampling were applied only to training data. This design follows the paper's stated intention that augmentation and scaling occur only on training data [1]. The five reproduced classifiers were KNN, Random Forest, Gradient Boosting, CatBoost and XGBoost. The final hold-out comparison used the parameter settings below. The paper specifies 10-fold cross-validation for validation; the submitted pipeline supports fold-wise leakage-safe SMOTE, but the final quantitative table in this report is based on the common stratified 80/20 hold-out because a complete 10-fold run across all five models was computationally prohibitive in the available environment. This is acknowledged as a limitation rather than presented as an exact replication of the paper's validation stage.

Model	Reproduction settings
KNN	<code>n_neighbors = 5</code>
Random Forest	200 trees, <code>random_state = 42</code> , parallel execution
Gradient Boosting	50 estimators, <code>learning_rate = 0.1</code> , <code>max_depth = 2</code>
CatBoost	200 iterations, <code>learning_rate = 0.05</code> , <code>depth = 6</code>
XGBoost	200 estimators, <code>learning_rate = 0.05</code> , <code>max_depth = 6</code> , <code>subsample = 0.8</code> , <code>colsample_bytree = 0.8</code>

Evaluation used Accuracy, weighted Precision, weighted Recall, weighted F1 and weighted one-vs-rest multiclass ROC-AUC. Macro-F1 and per-class metrics were also inspected when diagnosing minority-class behaviour.

2.6 Reproduced results

Model	Accuracy	Precision	Recall	F1	AUC
XGBoost	69.64%	67.50%	69.64%	68.14%	0.8455
Random Forest	66.85%	66.36%	66.85%	66.55%	0.8366
CatBoost	63.76%	66.19%	63.76%	64.72%	0.8361
Gradient Boosting	59.49%	64.89%	59.49%	61.48%	0.8224
KNN	40.35%	57.95%	40.35%	43.97%	0.7041

XGBoost was the strongest reproduced model, followed by Random Forest. This preserves the broad performance trend reported in the original study, in which XGBoost was also the strongest overall classifier [1]. However, all reproduced accuracies were substantially lower than the published values. Minority-class behaviour was particularly weak for HYP. In the reproduced XGBoost model, HYP precision was 0.17, recall 0.07 and F1 0.10, despite the use of SMOTE.

2.7 Comparison with the published results

Model	Published accuracy	Reproduced accuracy	Difference
XGBoost	93.00%	69.64%	-23.36 pp
Random Forest	91.00%	66.85%	-24.15 pp
CatBoost	89.00%	63.76%	-25.24 pp
Gradient Boosting	88.00%	59.49%	-28.51 pp
KNN	82.00%	40.35%	-41.65 pp

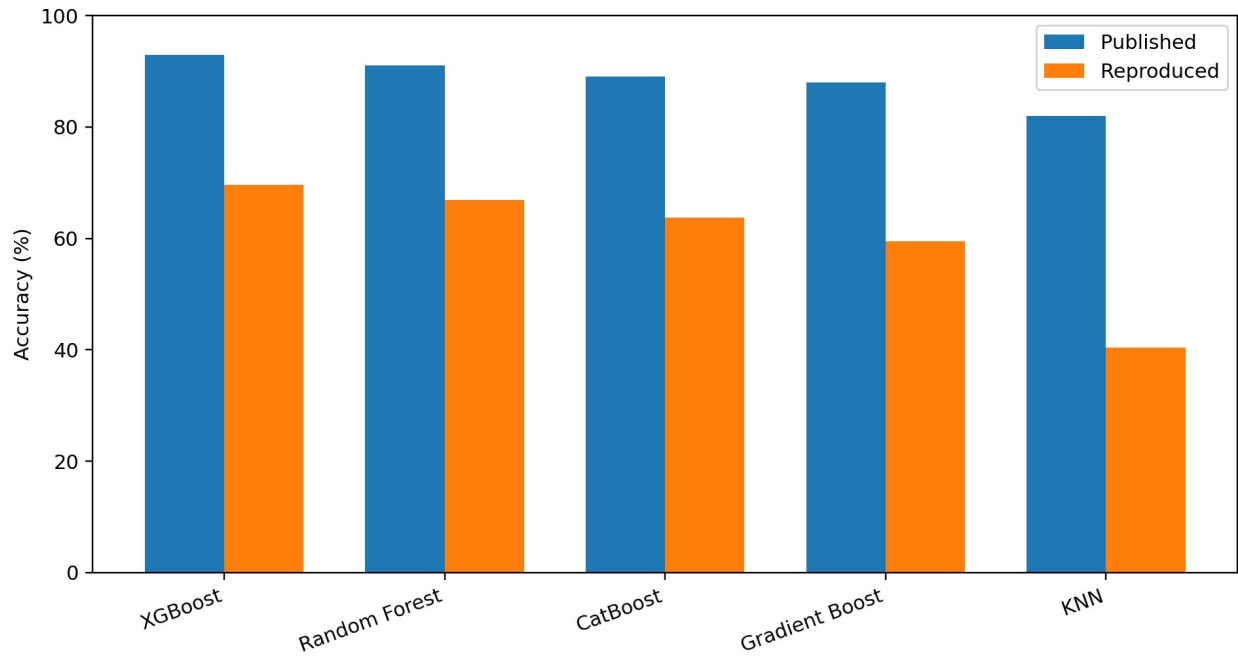


Figure 1. Published versus reproduced classifier accuracy.

The paper reports XGBoost accuracy, precision, recall and F1 of 93%, 92%, 90% and 93%, respectively, and reports class-wise XGBoost AUC values between 0.98 and 1.00 [1]. A direct AUC comparison is limited because the paper emphasises class-wise AUC, whereas this reproduction reports a weighted multiclass one-vs-rest AUC. The reproduced accuracy gap therefore should not be interpreted as a single implementation error; it reflects multiple methodological ambiguities and dataset differences.

2.8 Critical analysis of discrepancies

The largest source of uncertainty is the paper's feature representation. It clearly states that 12-lead ECG is used, but it does not describe how features across the twelve leads are merged before classification. Using Lead II makes the reproduction explicit and repeatable, but it may discard diagnostic morphology visible in other leads. A 12-lead extraction pilot on 1,500 ECGs was technically successful, with most recordings yielding all 12 usable leads, but a full multi-lead classification experiment was not used in the final baseline because the lead-aggregation rule in the paper remained unspecified.

A second source of variation is feature-length equalisation. The publication states that extracted feature lists have unequal lengths and are "equated" before flattening, but it does not report the target number of peaks or the exact padding/truncation operation [1]. The 11-beat representation used here is evidence-driven because 11 was the median valid beat count, but a different choice could materially alter the model input.

A third issue is dataset version and filtering. The paper reports 21,837 ECGs, while the reproduction metadata contained 21,799 records. After enforcing single-superclass labels and successful signal delineation, the modelling dataset contained 16,167 ECGs. The final class distribution was highly imbalanced, with only 535 HYP records compared with 9,067 NORM records. SMOTE can rebalance training counts, but it cannot replace information that is absent from the representation; this likely contributes to poor minority-class recall.

Finally, the paper does not provide full classifier hyperparameters. Reproduction therefore requires implementation choices that can affect absolute performance. The final results are consequently best interpreted as a transparent, independently executable reconstruction that preserves the paper's main comparative trend while exposing the implementation details that limit exact replication.

3. Part 2 - Proposed Machine Learning Solution

3.1 Limitation and research motivation

The Part 1 baseline depends on a fixed sequence of 11 beats. Longer ECGs lose later valid beats through truncation, while shorter ECGs are padded by repeating the last beat. This creates position dependence and may over-represent individual beats. It also produces 176 predictors even though many adjacent beat-level measurements are correlated. The proposed solution therefore redesigns the feature representation rather than merely replacing the classifier, satisfying the requirement for a methodological contribution.

The research question for Part 2 is: can a compact statistical representation of all valid beats reduce dimensionality and computational cost while retaining comparable predictive discrimination? This is also consistent with the original paper's emphasis on lightweight ECG processing for smart-healthcare deployment [1].

3.2 Proposed robust statistical feature representation

The proposed pipeline starts from the same validated beat-level PQRST and interval features used in Part 1. Instead of truncating or padding every ECG to 11 beats, all valid beats are retained. For each of the sixteen feature types, six distributional descriptors are calculated: mean, median, standard deviation, minimum, maximum and interquartile range. Two additional ECG-level descriptors are added: valid beat count and mean heart rate derived from RR intervals. This creates 98 predictors per ECG, compared with 176 in the baseline representation.

Using the same underlying extracted beats ensures that the comparison isolates the effect of the representation itself. The same 80/20 stratified split, StandardScaler, training-only SMOTE and XGBoost configuration were used in a controlled efficiency experiment. Training and prediction times were measured with the same hardware and Python environment, and the fitted pipelines were serialised to compare model file size.

3.3 Controlled experimental results

Measure	Baseline sequential	Proposed robust	Change
Number of features	176	98	-44.32%
Accuracy	0.6964	0.6892	-0.0071
Precision	0.6750	0.6721	-0.0029
Recall	0.6964	0.6892	-0.0071
Weighted F1	0.6814	0.6769	-0.0044
Macro F1	0.5104	0.5179	+0.0075
Weighted AUC	0.8455	0.8443	-0.0012
Training time	21.33 s	10.25 s	-51.94%
Prediction time	0.0351 s	0.0265 s	-24.43%
Model size	6.53 MB	5.38 MB	-17.63%

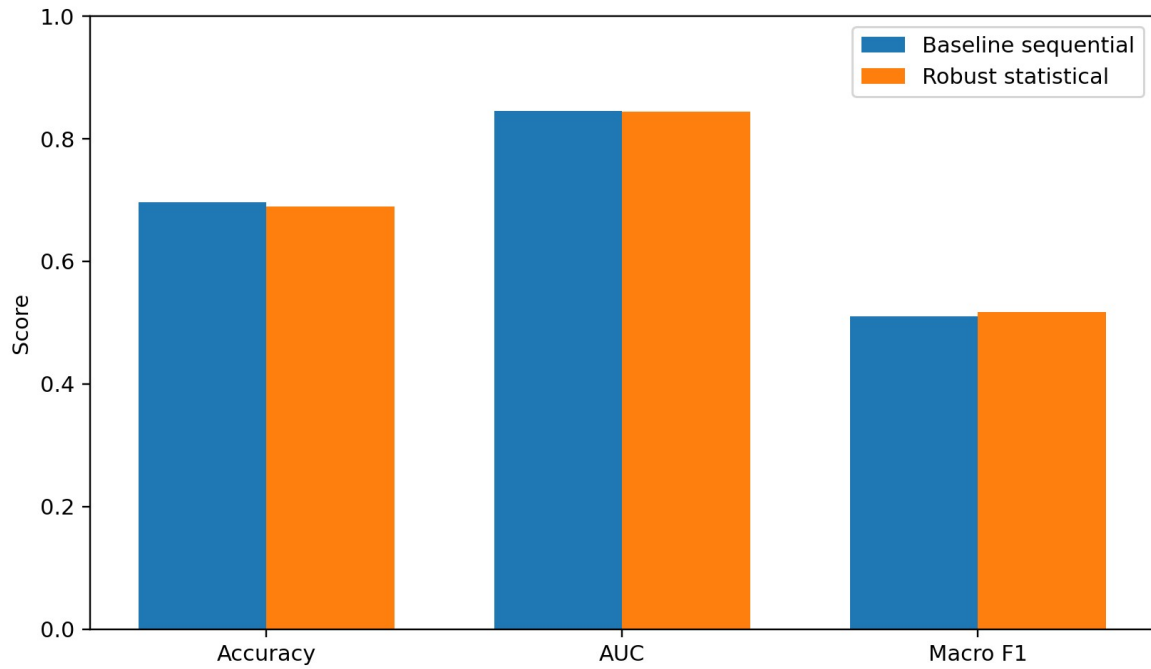


Figure 2. Predictive comparison of baseline and robust statistical representations.

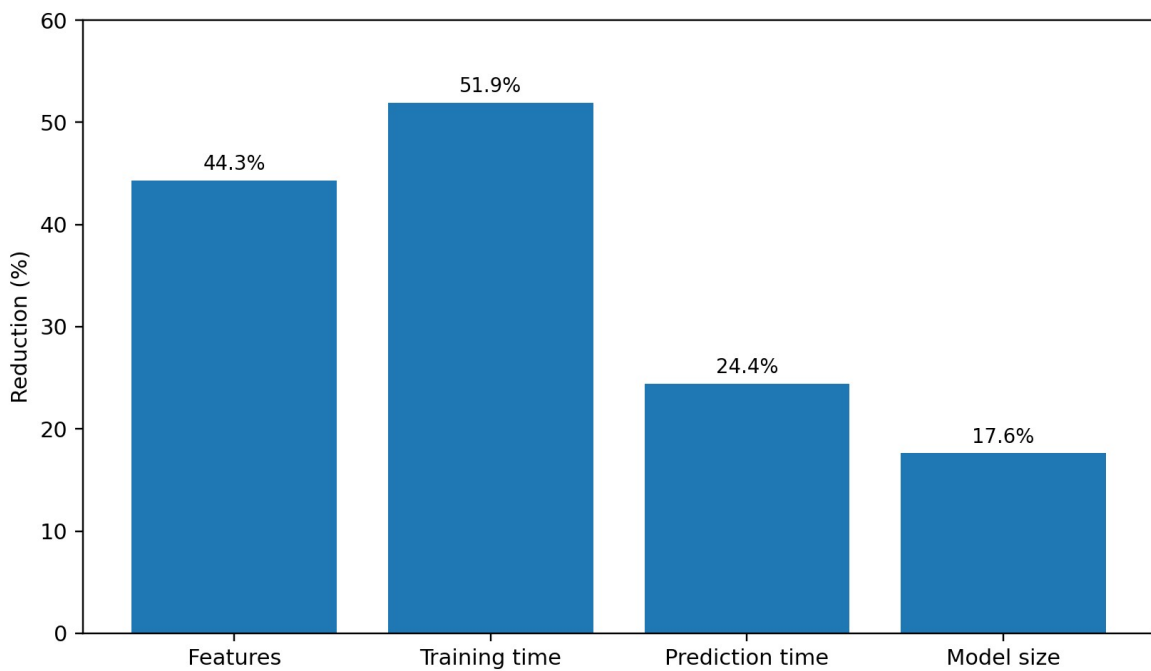


Figure 3. Computational and dimensionality reductions achieved by the proposed representation.

The proposed representation did not increase overall accuracy. Accuracy decreased by 0.71 percentage points and weighted F1 decreased by 0.44 percentage points. However, weighted AUC changed by only -0.0012, while macro-F1 improved from 0.5104 to 0.5179. The main improvement is efficiency: the feature space was reduced by 44.32%, training time by 51.94%, prediction time by 24.43%, and model size by approximately 17.63%. These results support a claim of improved compactness and computational efficiency with comparable discrimination, not a claim of superior raw predictive accuracy.

3.4 Additional exploratory experiments

Two additional strategies were tested to determine whether the Part 1 baseline could be improved directly. A Random Forest + XGBoost stacking ensemble achieved 67.75% accuracy and 0.8181 AUC, which was lower than the XGBoost baseline. A fused representation combining the sequential and robust statistical features produced 274 predictors and achieved 69.42% accuracy and 0.8467 AUC. The slight AUC increase (+0.0012) was accompanied by small decreases in accuracy and weighted F1, so the additional dimensionality was not justified. These negative results were retained because they show that simply adding models or features did not address the underlying representation and minority-class problems.

3.5 Strengths, limitations and practical implications

The proposed robust representation has three main strengths. It uses all valid beats instead of an arbitrary fixed sequence, substantially reduces input dimensionality, and cuts training and inference time while maintaining nearly identical AUC. These properties align with the selected paper's smart-healthcare motivation, where lighter models are easier to deploy and maintain [1].

The main limitation is that minority-class recognition remains weak and overall accuracy does not improve. Statistical summaries can remove beat-to-beat temporal structure that may be diagnostically useful, while Lead II alone may not contain all morphology required for HYP or other classes. Future work should therefore test explicit multi-lead aggregation, cost-sensitive learning, class-specific feature selection, patient-level grouped validation, and calibrated models. The original paper itself identifies reduced-lead and wearable ECG as future directions, which further motivates studying compact but information-preserving representations [1].

4. Discussion

The reproduction demonstrates the difference between reproducing a paper's high-level workflow and reproducing its exact numerical results. The major stages of DASMCC - PTB-XL preparation, single-label filtering, time-domain ECG feature extraction, SMOTE-based imbalance handling and comparison of five classifiers - were reproduced. XGBoost remained the strongest overall model, which is consistent with the selected paper. However, the absolute accuracy gap indicates that undocumented implementation details materially influence the published performance.

The most important learning from Part 2 is that "improvement" should be evaluated across multiple objectives. The robust representation did not improve accuracy, but it reduced the feature space almost by half and halved training time while leaving AUC essentially unchanged. In an edge or smart-healthcare setting, that trade-off can be practically valuable. At the same time, the results show that efficiency should not be used to hide poor minority-class behaviour; HYP remains a difficult class and requires further methodological attention.

A further methodological issue is validation. The paper reports 10-fold cross-validation, while the final reproducible comparison in this report uses a common stratified hold-out. Future work should complete patient-grouped or stratified-group cross-validation so that repeated ECGs from the same subject cannot influence both training and validation folds. This would provide a stricter estimate of generalisation and would be preferable to relying only on random record-level splitting.

5. Conclusion

This study reproduced the DASMCC machine-learning workflow for five-class cardiovascular-disease prediction using PTB-XL ECG data. After single-label filtering and validated beat extraction, 16,167 ECGs were used for modelling. XGBoost achieved the strongest reproduced performance with 69.64% accuracy, 68.14% weighted F1 and 0.8455 weighted AUC. Although these values were below the published 93% accuracy, the reproduction preserved the original ranking trend and revealed several reproducibility constraints, including unspecified 12-lead feature consolidation, ambiguous feature-length equalisation, dataset-version differences and missing hyperparameters.

The proposed Part 2 solution replaced the fixed 11-beat representation with robust statistical summaries of all valid beats. This reduced the input feature count by 44.32%, training time by 51.94% and prediction time by 24.43%, while weighted AUC remained effectively unchanged and macro-F1 improved slightly. The contribution is therefore a more

compact and computationally efficient representation rather than a higher-accuracy classifier. Future improvements should focus on multi-lead information, patient-level validation and methods specifically targeted at minority-class recognition.

6. Reflection

The most useful lesson from this task was that research reproduction requires more than rerunning an algorithm. Small decisions about labels, peak matching, sequence length, class balancing and data splitting can materially change results. The initial feature extractor produced negative ECG intervals, which revealed that the P, Q, R, S and T arrays could not simply be aligned by position. Enforcing valid beat ordering improved the reliability of the preprocessing pipeline and made the later modelling results easier to defend.

I also learned that a proposed improvement does not need to be judged by accuracy alone. The stacking and fused-feature experiments did not outperform the baseline, but they helped isolate the representation problem. The final robust representation provided a measurable efficiency improvement with nearly unchanged AUC. This changed my interpretation of the task from "trying to reach the published score" to designing and evaluating a reproducible method with clear evidence, limitations and practical trade-offs.

7. Part 3 - Video Presentation

Video link: [INSERT ACCESSIBLE YOUTUBE / PLATFORM LINK BEFORE SUBMISSION]

The video should be no longer than six minutes, show face and screen, demonstrate selected code execution and model outputs, explain the reproduction and proposed method, and reflect on limitations and learning. The video content must remain consistent with this report and the submitted code package.

8. GenAI Acknowledgement

Generative AI (ChatGPT) was used during this assessment to support planning and brainstorming of the experimental workflow, to discuss possible debugging approaches, and to assist with grammar and language editing. All datasets were downloaded and processed by me, all Python scripts were executed and checked in my local environment, and the reported numerical results come from those executions. I reviewed the generated suggestions, modified the implementation where required, and am responsible for the final interpretation, accuracy and integrity of the submission. This acknowledgement should be reviewed and adjusted before submission so that it accurately reflects my final use of GenAI in accordance with the unit assessment requirements.

9. Reproducibility Package

Code / ZIP / GitHub link: [INSERT ACCESSIBLE LINK BEFORE SUBMISSION]

The submitted package should contain the Python source files, at least one required dataset or an approved dataset subset, generated result CSV files, figures, environment/package instructions, and a README describing the commands required to reproduce the report results from start to finish.

File	Purpose
prepare_labels.py	Parse SCP statements and create the single-superclass dataset.
extract_features_full.py	Extract validated beat-level PQRST and interval features.
prepare_model_data.py	Equalise to 11 beats and create the 176-feature Part 1 dataset.
train_baseline_models.py	Train and evaluate the five Part 1 classifiers.
part2_robust_features.py	Create the 98-feature robust statistical representation.
part2_efficiency_comparison.py	Run the controlled baseline-vs-proposed efficiency comparison.

References

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Appendix A - Key Reproduction Statistics

Item	Value
PTB-XL metadata records used in reproduction	21,799
Single-superclass records before ECG extraction	16,244
Successfully processed ECGs	16,167
Failed ECGs	77
Feature-extraction success rate	99.53%
Total valid beat rows	176,302
Mean valid beats per ECG	10.91
Median valid beats per ECG	11
Part 1 predictors per ECG	176
Part 2 predictors per ECG	98